Detector production: 8,000-event ion-grid scan

NAUSICAA generated report

Configuration

Each of the two 1 mm Geant4 samples uses one shared event placement and one Fano-sampled primary-pair realisation. Its ions are collected on three pad grids. The electron branch uses the same realization and is propagated once per sample, producing separate pure-Xe and Xe–15% He incident-photon histories. The electron settings are identical for all ion-grid choices.

Parameter	Value
TPC length	150.0 cm
Pair-creation energy W	21.90 eV
Fano factor F	0.150
Electron lifetime	15.0 ms
Electron hole pitch / bucket	10.0 mm / 2.0 μs
EL E/P and J	3.0 kV/cm/bar; 0.010

The $J = 0.01$ operating-point input is the user's approximate reading of Oliveira et al., Figure 6. The saved histories are ions per pad/time frame and photons incident on each SiPM/time bucket. SiPM PDE and electronics are subsequent stages.

Event and charge ledgers

Sample	Events	Pairs/ions	Surviving e^-	Xe photons	Xe–He photons
bb0nu	8,000	897,306,359	856,854,917	381,687,642,280	375,319,166,815
single electrons	8,000	897,244,198	856,968,162	381,739,047,046	375,368,751,796

Every ion-grid file has exactly one collected ion per sampled pair. Each ion-only replay verified every event's pair checksum against the shared realization before collecting charge.

Ion-grid outputs

Sample	Grid pitch	Frame	Ions	File size	Ion-only time
bb0nu	2 mm	20 ms	897,306,359	138.8 MB	45.6 s
bb0nu	3 mm	30 ms	897,306,359	125.3 MB	joint run
bb0nu	4 mm	40 ms	897,306,359	120.4 MB	42.8 s
single electrons	2 mm	20 ms	897,244,198	142.9 MB	43.5 s
single electrons	3 mm	30 ms	897,244,198	127.3 MB	joint run
single electrons	4 mm	40 ms	897,244,198	121.7 MB	40.1 s

The 3 mm histories were produced in the joint ion/electron run; their separate ion-only runtime was not measured. The index HDF5 stores full paths and source/configuration identities for every branch.

Provenance

Run ID: `238ddc5a75e0fd9b`. Numerical index: `grid_scan_summary_238ddc5a75e0fd9b.h5`. The index records the SHA-256 digest of the joint-run summary and the resolved realisation, ion and electron TOML cards.